

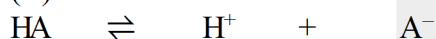
HINTS & SOLUTIONS

Q.1 (A)

Molar conductance of HCl will be maximum and CH₃COOH will be minimum because

- [H⁺] have exceptional ionic mobility
- CH₃COOH is a weak electrolyte

Q.2 (C)



$$C(1-\alpha)$$

$$C\alpha$$

$$C\alpha$$

$$\alpha = \frac{\Lambda_{\text{eq}}}{\Lambda_{\text{eq}}^{\infty}} \Rightarrow 10/200 = 0.05$$

$$[\text{H}^+] = 0.1 \times 0.05 \Rightarrow 0.005$$

$$\text{pH} = 3 - \log 5 \Rightarrow 2.3$$

Q.3 (B)

Q.4 (C)

Q.5 (B)

$$E = (-1.05 + 10^{-4} T^2) \text{V}$$

(A) $\left[\frac{\partial E}{\partial T} \right]_P = 0 + 10^{-4} \times 2T$

$$\Delta H^{\circ} = -nFE^{\circ} + nFT \left[\frac{\partial E}{\partial T} \right]_P$$

ΔH° will depends on temperature

(B) $\Delta S = nF \left[\frac{\partial E}{\partial T} \right]_P > 0$ Thus entropy will increase.

(C) $\Delta G = -nFE$ as $E = -1.05 + 10^{-4} T^2$

It can be less than zero at certain temperature so reaction will not be spontaneous at all temperature.

(D) At 200 K

$$E = -1.05 + 10^{-4} \times (200\text{K})^2$$

$$= -1.05 + 0.02$$

$$E = -1.03 \text{V}$$

thus reaction is non-spontaneous.

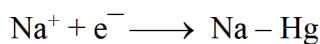
Q.6 (D)

Q.7 (A)

Q.8 (C)

$$\left(\frac{x}{10+x} \right) \times 100 = 18.7$$

$$x = 2.3 \text{ or } 0.1 \text{ mol}$$



$$= 0.2 F (\eta = 0.5)$$

Q.9 (C)

Q.10 (B)

Q.11 (C)

Q.12 (B)

Q.13 (B)

Q.14 (A)

$$(i) \quad [\text{Ag}^+]_{\text{added}} = \frac{0.1}{0.250} = \frac{2}{5} = 0.4$$

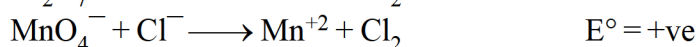
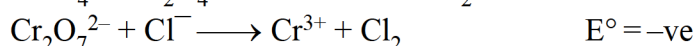
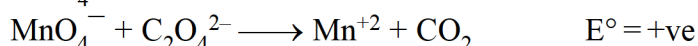
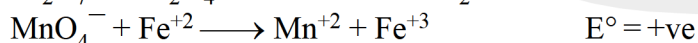
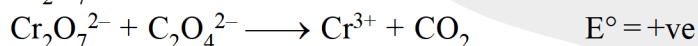
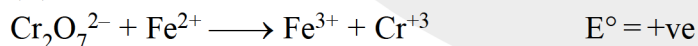
$$K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-]_{\text{left}} \\ 10^{-10} = 0.4 \times [\text{Cl}^-]_{\text{left}} \\ [\text{Cl}^-]_{\text{left}} = 2.5 \times 10^{-10} \text{ M}$$

(ii) Since $[\text{Cl}^-]$ is negligible, hence at cathode Ag^+ and at anode O_2 will be liberated.

$$(iii) \quad [\text{Ag}^+] = 0.4 \text{ mole/L} \quad [\text{NO}_3^-] = 0.4 \text{ mole/L} \quad [\text{Cl}^-] = 2.5 \times 10^{-10}$$

$$\begin{aligned} \text{Total Conductivity} &= \frac{60 \times 0.4}{1000} + \frac{75 \times 0.4}{1000} + \frac{2.5 \times 10^{-10} \times 75}{1000} \\ &\quad \text{(due to Ag}^+\text{)} \quad \quad \quad \text{(Due to NO}_3^-\text{)} \quad \quad \quad \text{(Due to Cl}^-\text{)} \\ &= 5.4 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1} \end{aligned}$$

Q.15 (C)



$\therefore \text{MnO}_4^-$ can't be used to estimate FeC_2O_4 in presence of dil.HCl as it will also oxidise Cl^- to Cl_2 .

Q.16 (B)

$$\text{Millimoles of KMnO}_4 \text{ used} = 50 \times 0.1 = 5$$

$$M_{\text{eq}} \text{ of KMnO}_4 \text{ used} = 5 \times 5 = 25 = M_{\text{eq}} \text{ of Cl}^- + M_{\text{eq}} \text{ of FeC}_2\text{O}_4$$
$$(n_f = 5)$$

$$\therefore M_{\text{eq}} \text{ of Cl}^- = M_{\text{eq}} \text{ of Cl}_2 = 3.5 \times 2 = 7$$
$$(n_f = 2)$$

$$\therefore M_{\text{eq}} \text{ of FeC}_2\text{O}_4 = 25 - 7 = 18$$
$$(n_f = 3)$$

$$\therefore \text{Millimoles of FeC}_2\text{O}_4 = \frac{18}{3} = 6$$

Q.17 (B)

Q.18 (C)

Q.19 (A)

Q.20 (B)(C)

Q.21 (A) P, Q (B) P, Q (C) Q, R, (D) P, S

Q.22 1.39V

Q.23 43.456g

Q.24 0.372 V, -0.307 V

Q.25 0.522 V, 0.613 V

Q.26 121.6gm, 76gm

$$\text{Equivalent of Ag} = \frac{i \times t}{96500} = \frac{2 \times 9650}{96500} = 0.2$$

$$\frac{W_{\text{Ag}}}{108} = 0.2$$

$$W_{\text{Ag}} \text{ deposit} = 0.2 \times 108 = 21.6 \text{ gram}$$

100 gm cathode will gain 21.6 gm Ag its wt. will become 121.6 gm.

In order to release 21.6 gm Ag from anode

$$\text{Weight lose anode} \times \frac{90}{100} = 21.6 \text{ gm Ans.}$$

$$\text{Weight lose anode} = 24 \text{ gm}$$

$$\text{weight anode after electrolysis} = 100 - 24 = 76 \text{ gm}$$

Q.27 270

$$\frac{y-260}{x-\sqrt{0.25}} = \frac{260-250}{\sqrt{0.25}-1}$$

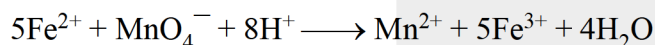
$$y-260 = (-20) \left(x - \frac{1}{2} \right)$$

$$y = 270 - 20x$$

$$\text{Intercept} = 270$$

$$\wedge_m^\infty = 270 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

Q.28 0394



$$\Delta G^\circ = -5 \times F \times E_{\text{cell}}^\circ$$

$$= -5 \times F \times 0.8 \text{ Joule}$$

$$\Delta G^\circ = \Delta G_{\text{f Mn}^{2+}}^\circ + 5\Delta G_{\text{f Fe}^{3+}}^\circ + 4\Delta G_{\text{f H}_2\text{O}(l)}^\circ - 5\Delta G_{\text{f Fe}^{2+}}^\circ - \Delta G_{\text{f MnO}_4^-}^\circ - 8\Delta G_{\text{f H}^+}^\circ$$

$$-5 \times 96500 \times 0.8 \times 10^{-3} = (-229) + 5(-10.2) + 4(-230) - 5(-84) - \Delta G_{\text{f MnO}_4^-}^\circ$$

$$\Delta G_{\text{f MnO}_4^-}^\circ = -780 + (5 \times 96500 \times 0.8 \times 10^{-3})$$

$$= -394$$

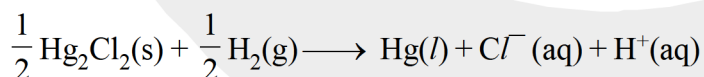
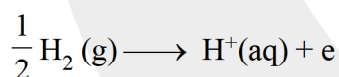
Ans. 0394

Q.29 6

At cathode



At anode



$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{1} \log [\text{Cl}^-][\text{H}^+] \quad \text{or,} \quad 0.634 = (0.28 - 0) + 0.059 \text{ pH}$$

$$\text{or, pH} = \frac{0.634 - 0.28}{0.059} = 6$$

Q.30 $-1.30 \times 10^3 \text{ kJ mol}^{-1}$

